Climate Trend and Weather Prediction with State-Space Models

Time Series Analysis (20236) - Group 10

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Scientific question

This project studies two related climate time-series questions. First, using GISTEMP, we ask whether Northern Hemisphere temperature anomalies show a persistent long-run increase, and whether this increase can be represented either as a sequence of discrete regimes or as a continuously evolving latent level. Second, using GHCN daily data, we ask whether modelling dependence between nearby weather stations improves one-step-ahead prediction of maximum temperature. We address these questions with Hidden Markov Models and Dynamic Linear Models: HMMs provide a regime-based description of long-run warming, while DLMs provide a state-space framework for latent trend estimation and multivariate weather prediction.

1. GISTEMP data exploration

The GISTEMP data contain monthly temperature anomalies relative to the 1951–1980 reference period. The monthly observations were arranged as a regular time series from January 1880 to December 2025. The monthly series used in the analysis contains 0 missing values.

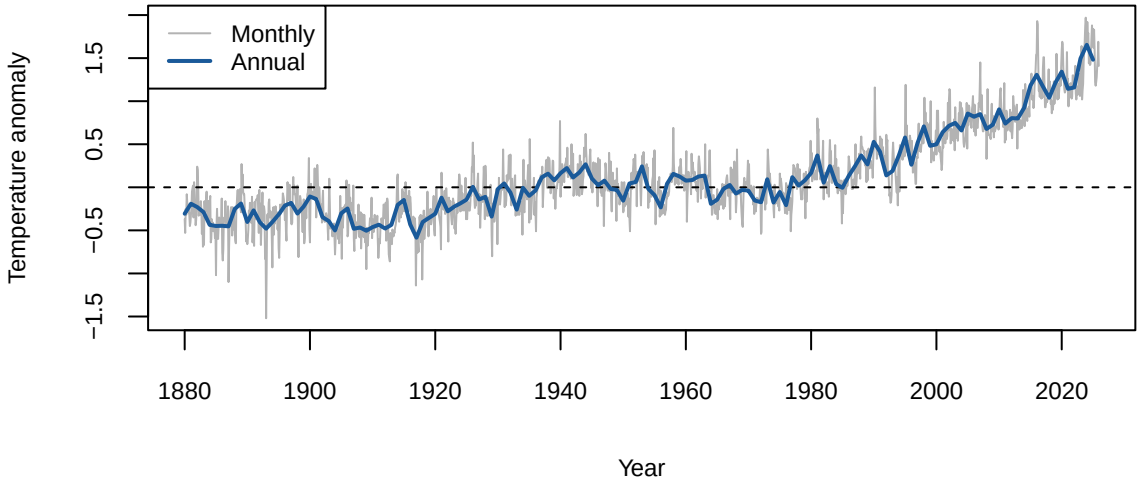


Figure 1. Monthly GISTEMP temperature anomalies (gray) and annual averages (blue), 1880–2025.

Figure 1 shows strong monthly fluctuations around the annual average, but the low-frequency pattern is clearly upward. The annual average is useful because it filters out high-frequency monthly noise without changing the long-run scale of the data.

Table 1. Mean and standard deviation of annual temperature anomalies by period.

period	Mean annual anomaly	SD annual anomaly
1880-1950	-0.209	0.210
1951-1980	0.000	0.127
1981-2025	0.699	0.426

Table 1 gives a quantitative summary of this pattern. The reference-period mean is approximately zero by construction, while the recent period has both a much higher mean anomaly and a larger standard deviation. The mean annual anomaly increases from -0.209 in 1880–1950 to 0.699 in 1981–2025 (difference: 0.908), consistent with a persistent upward shift rather than only a few extreme months.

2. Decomposition

We use the additive decomposition

$$Y_t = T_t + S_t + e_t,$$

where T_t is the trend component, S_t is the seasonal component, and e_t is the irregular component. Since temperature anomalies can be negative, an additive decomposition is more appropriate than a multiplicative one.

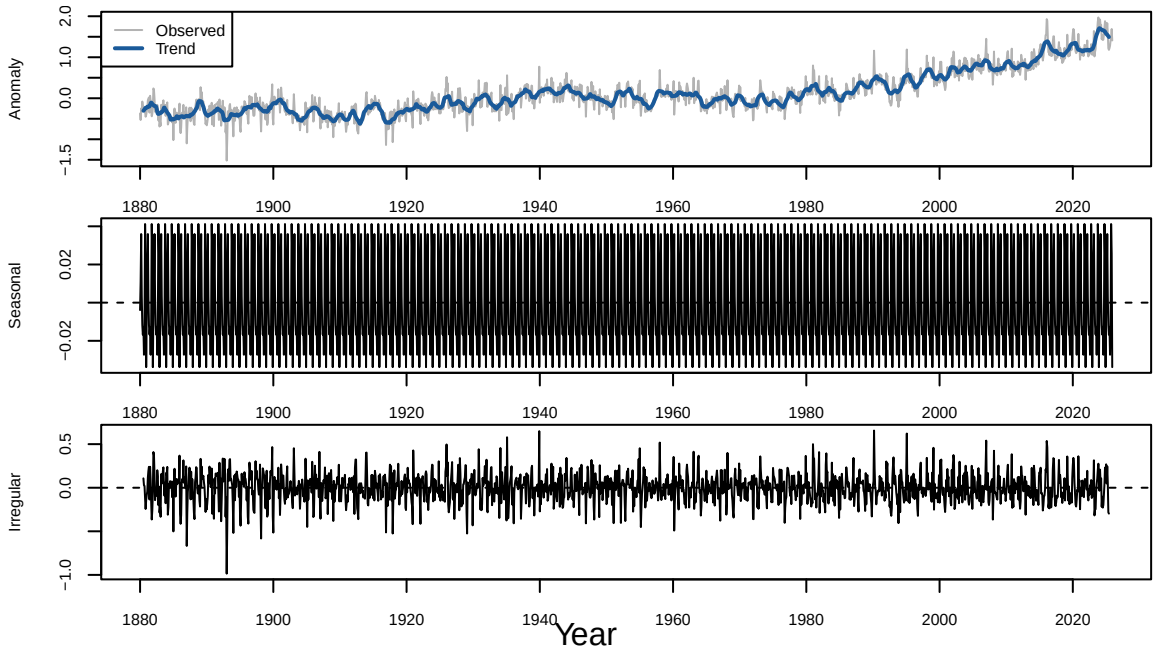


Figure 2. Additive decomposition of monthly GISTEMP anomalies: observed and trend (top), seasonal component (middle), irregular component (bottom).

Figure 2 confirms that the trend component dominates the decomposition. The seasonal component is small relative to the trend, which is expected because the series is already expressed as temperature anomalies rather than raw temperatures. Removing the estimated seasonal component therefore changes the series only slightly, but it keeps the following HMM and DLM analyses focused on regimes and the latent warming trend rather than residual monthly seasonality.

3. Do the data document global warming?

We now work with the seasonally adjusted monthly GISTEMP series and study whether the long-run increase can be represented through discrete regimes or through a smooth latent level.

3.1 Hidden Markov Model

We use a Gaussian Hidden Markov Model. Let Z_t be an unobserved regime indicator:

$$P(Z_t = j \mid Z_{t-1} = i) = p_{ij}, \qquad Y_t \mid Z_t = k \sim N(\mu_k, \sigma_k^2).$$

The regimes are interpreted as different temperature-anomaly levels. This is a simple way to summarize long-run changes: instead of imposing a deterministic trend, the HMM lets the data switch between persistent states with different mean anomalies. Since the underlying process is trending, the regimes should be read as descriptive states rather than literal physical climate regimes.

In this application, the HMM is used mainly as an exploratory regime-switching model. The latent state Z_t is not observed directly, but it summarizes the position of the temperature anomaly series in a low, intermediate, or high-temperature regime. Conditional on the state, the observations are assumed Gaussian with regime-specific mean and variance. The transition matrix controls the persistence of each regime: large diagonal probabilities indicate that the process tends to remain in the same regime for several consecutive months.

The main advantage of this approach is interpretability. A continuous and noisy series is converted into a small number of persistent states, making the long-run shift easier to visualize. This is particularly useful for detecting whether high-temperature observations are isolated events or whether they form a persistent regime. However, the HMM also has an important limitation in this setting: global warming is plausibly gradual, while the HMM forces the process into a finite number of discrete states. Therefore, the estimated regimes should not be interpreted as physical climate states, but as a statistical summary of the changing distribution of anomalies.

Table 2. HMM model comparison.

Model	Log-likelihood	AIC
2 regimes	-430.0	874.0
3 regimes	-32.2	92.4

Table 3. Estimated HMM regime-specific parameters.

Regime	Mean anomaly	SD anomaly	Decoded share
Cooler	-0.302	0.218	0.416
Intermediate	0.113	0.186	0.366
Warmer	0.907	0.361	0.218

Table 4. Estimated HMM transition matrix.

From	Cooler	Intermediate	Warmer
Cooler	0.977	0.023	0.000
Intermediate	0.024	0.969	0.007
Warmer	0.000	0.009	0.991

Tables 2–4 show that the three-regime model is preferred by AIC. The estimated means order the regimes naturally as cooler, intermediate, and warmer anomaly states. The transition matrix has large diagonal entries, meaning that the decoded states are persistent. The warmest-regime persistence is 0.991, but this should not be interpreted as a truly absorbing climate state: it mainly reflects fitting a finite-state model to a strongly trending series.

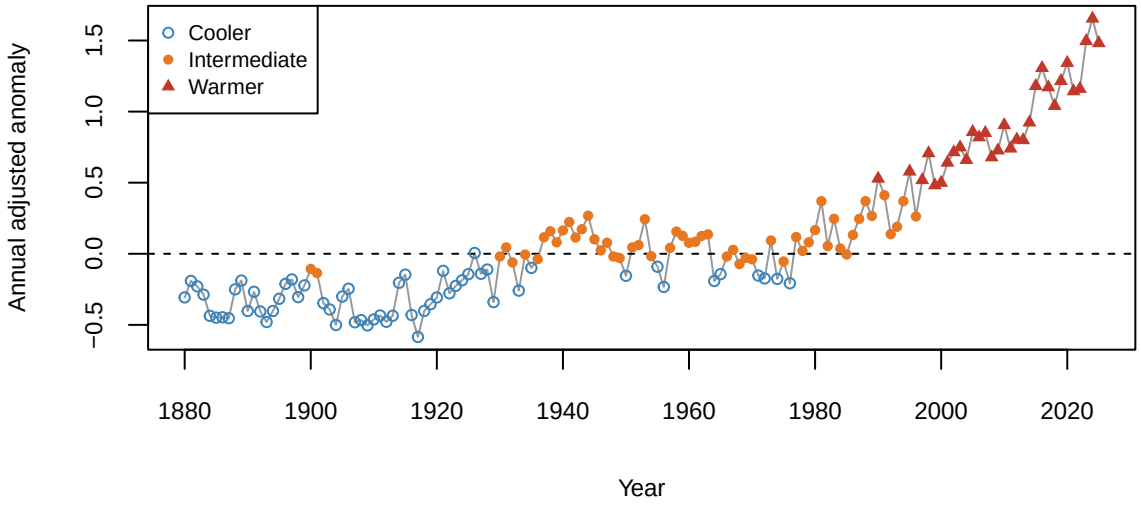


Figure 3. Annual seasonally adjusted anomalies, colour-coded by the dominant HMM regime decoded for each year.

Figure 3 shows a transition from cooler to warmer regimes. The warmest regime appears first around 1981; its share is 0 before 1951 and 0.707 after 1980. The HMM therefore summarizes warming as a persistent movement from low-anomaly states to high-anomaly states, but it necessarily discretizes a process that may be gradual.

3.2 Dynamic Linear Model

The benchmark model is a random walk plus noise model,

$$Y_t = \mu_t + v_t, \quad v_t \sim N(0, V), \qquad \mu_t = \mu_{t-1} + w_t, \quad w_t \sim N(0, W).$$

We also estimate two local linear trend specifications as sensitivity checks, since those models contain an explicit local slope. However, the preferred DLM is selected using AIC. In this data set, the random walk plus noise model gives the best likelihood-penalized fit, so we use it to estimate the latent level. Acceleration is summarized by the local yearly change of the selected smoothed level, $\mu_t - \mu_{t-12}$. This is not a separate stochastic slope; it is a derived measure of how much the estimated latent level changes over one year.

Table 5. Comparison of alternative DLM specifications.

Model	Log-likelihood	AIC	Selected
Random walk plus noise	2039.28	-4074.55	Yes
Unrestricted local linear trend	2025.11	-4044.22	No
Smooth local linear trend	1894.52	-3785.03	No

The random walk plus noise model has the lowest AIC and is selected. The local linear trend alternatives are useful robustness checks, but they do not improve the fit in this application. The Kalman smoother gives a retrospective estimate of the latent level, using the full sample. We therefore interpret it as a historical estimate of the long-run anomaly level rather than as a real-time forecast. Acceleration is measured through twelve-month changes in that smoothed level.

Table 6. Estimated variance parameters of the selected DLM.

Quantity	Estimate
Observation variance V	0.020450
Level innovation variance W	0.006367

Table 7. Average local yearly change in the smoothed DLM level by period.

period	Mean yearly change
1880-1950	0.0021
1951-1980	0.0121
1981-2025	0.0283

Table 7 shows that the average local yearly change rises from 0.0021 in 1880–1950 to 0.0283 in 1981–2025, consistent with a faster recent increase.

Table 8. Basic diagnostics for the selected DLM.

Quantity	Value
Log-likelihood	2039.28
AIC	-4074.55
SD standardized residuals	1
Ljung-Box p-value, lag 24	< 0.001

Table 8 shows that the standardized residuals are correctly scaled, but the Ljung-Box p-value indicates remaining monthly autocorrelation. This is not surprising for climate data, where short-run dependence

can remain even after smoothing. For this reason, the DLM is interpreted mainly as a model for the latent long-run level and its local yearly change, not as a complete model for all monthly dependence.

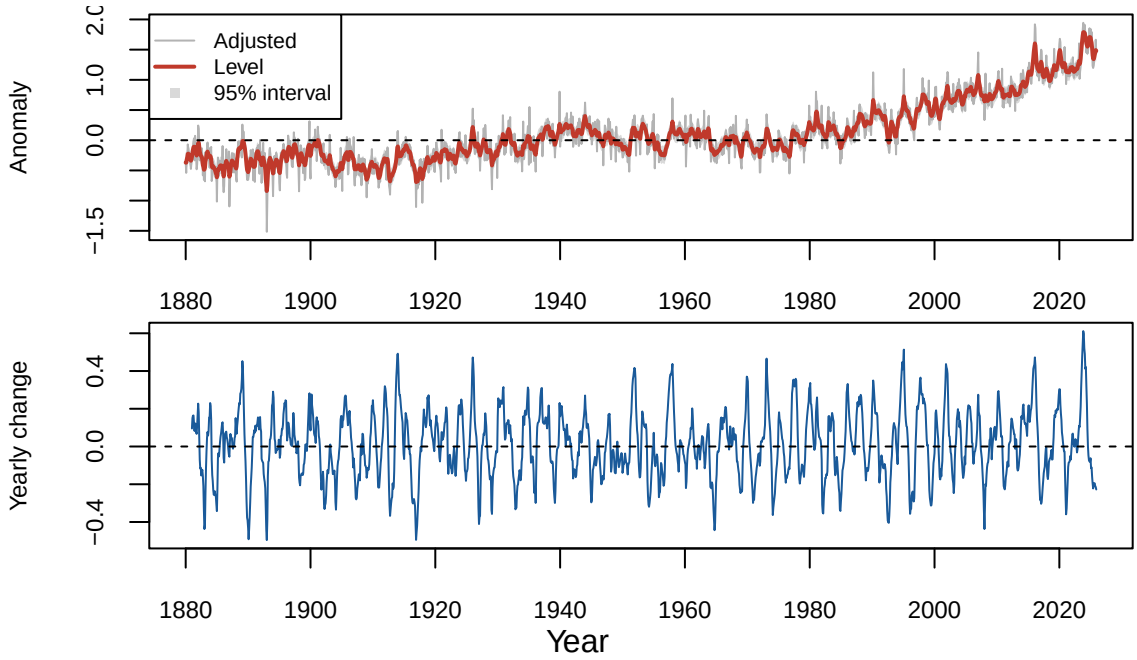


Figure 4. DLM results: smoothed latent level (top) and local yearly change in the smoothed level (bottom).

Figure 4 shows a rising smoothed level (final value 1.481). Because yearly changes are volatile, acceleration is read through period averages: the mean rises from 0.0021 in 1880–1950 to 0.0283 in 1981–2025.

3.3 Comparison of HMM and DLM results

Both models support long-run warming but describe it differently. The HMM represents the series as switching between persistent anomaly regimes, which is useful for visualizing changes in level or possible change points. The DLM instead treats warming as a continuously evolving latent process, and the local yearly change provides a simple measure of the speed of that evolution. Their limitations are complementary: the HMM discretizes a gradual process, while the DLM leaves some short-run autocorrelation unexplained.

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4. GHCN Weather Prediction

We now study daily maximum temperatures at two nearby GHCN stations in Maine. This part of the project focuses on short-run weather prediction rather than long-run climate trends. The goal is to model the joint dynamics of the two stations with bivariate DLMs of increasing spatial complexity and to assess whether cross-station structure improves one-step-ahead prediction.

4.1 Data Preparation

We select **Houlton AP** and **Augusta State AP**, two geographically close stations whose common records span 1948-06-01 to 2024-12-31, yielding 27,029 complete daily observations after aligning the two stations and removing missing TMAX values. TMAX is converted from tenths of degrees Celsius. Seasonality is removed by subtracting, for each station and calendar day, the training-sample day-of-year mean; the same seasonal means are then applied to the test period, avoiding leakage. The adjusted series can be interpreted as daily deviations from the typical seasonal cycle.

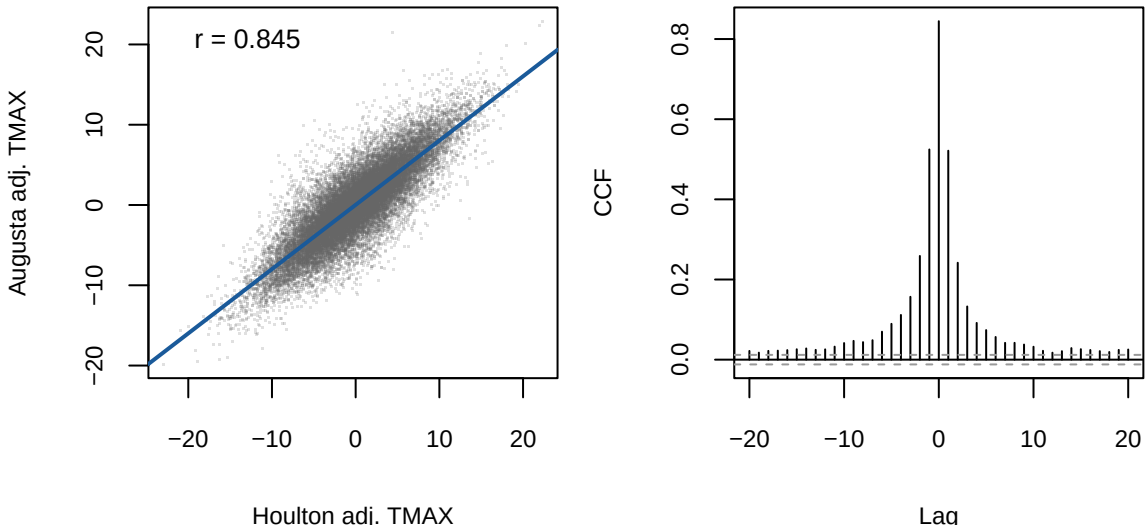


Figure 5. Dependence between adjusted TMAX at Houlton and Augusta: scatter plot and cross-correlation function.

Figure 5 shows strong contemporaneous dependence (correlation 0.845) and a CCF peaking at lag 0, suggesting synchronous regional co-movement. This motivates bivariate state-space models: if both stations are driven by the same regional weather shocks, allowing correlated state innovations should improve the description of the joint dynamics.

4.2 Model Specification and Estimation

Let $Y_t = (Y_{1,t}, Y_{2,t})^\top$ denote adjusted TMAX at Houlton and Augusta. The three required bivariate DLMs share

$$Y_t = \theta_t + v_t, \quad v_t \sim N(0, V), \qquad \theta_t = \theta_{t-1} + w_t, \quad w_t \sim N(0, W),$$

and differ only in the covariance structure. Model 1 assumes diagonal V and W, so the stations evolve independently. Model 2 keeps V diagonal but estimates

$$W = \begin{pmatrix} \sigma^2_{w,1} & \sigma_{w,12} \\ \sigma_{w,12} & \sigma^2_{w,2} \end{pmatrix},$$

allowing correlated latent innovations. Model 3 replaces this unrestricted matrix with the spatial restriction

$$W = \sigma^2_w \begin{pmatrix} 1 & \rho_{12} \\ \rho_{12} & 1 \end{pmatrix}, \qquad \rho_{12} = \exp(-\phi d_{12}),$$

where $d_{12} = 2.695$ is the Euclidean distance in latitude–longitude degrees between the two station coordinates. Thus Model 2 is the most flexible dependence model, while Model 3 is the geographically interpretable version.

Table 9. Maximum likelihood estimates for the three bivariate DLMs.

Parameter	Model 1	Model 2	Model 3
Obs. SD Houlton	2.8615	1.8626	1.9917
Obs. SD Augusta	2.7743	1.8262	1.6509
State SD Houlton	2.8043	3.9376	3.8068
State SD Augusta	2.5734	3.6909	3.8068
State corr.	0	0.999	0.9975
Spatial phi	–	–	0.0009

Table 9 shows that Models 2–3 estimate near-perfect state-innovation correlation, consistent with a strong regional temperature signal shared by the two nearby stations. Model 2 obtains this through an unrestricted covariance matrix, while Model 3 obtains it through a spatial decay parameter close to zero, implying nearly flat dependence over this short distance.

The estimated standard deviations indicate that both observation noise and latent state innovations contribute substantially to daily temperature variability. In Models 2 and 3, the state innovation standard deviations are large and highly correlated across stations, suggesting that a sizeable part of daily variation is explained by common regional shocks rather than by independent station-specific noise. At the same time, the residual diagnostics show that the random-walk state equation does not fully capture short-run persistence, which motivates the VAR(1) benchmark and richer state-space extensions.

4.3 Model Comparison and Predictive Performance

Parameters are estimated on 1948–2022 only (26298 days). During 2023–2024, one-step-ahead predictions use observations up to the previous day, with parameters fixed at their training estimates. Thus RMSE values below measure genuine out-of-sample predictive accuracy. Log-likelihood and AIC are reported for the three required DLMs. VAR(1) is an autoregressive benchmark, while Model 4 is an optional spatial seasonal DLM fitted to raw TMAX, adding harmonic seasonal states so that seasonality is modelled internally rather than pre-adjusted.

Table 10. Predictive performance for the three DLMs, a VAR(1) benchmark, and optional Model 4. RMSE is evaluated on 2023–2024; log-likelihood and AIC are reported only for the three required DLMs.

Model	RMSE Houlton	RMSE Augusta	Log-likelihood	AIC	Role
Model 1	4.2347	4.1759	-105002.8	210013.6	DLM
Model 2	4.2474	4.0691	-90230.7	180471.4	DLM
Model 3	4.2196	4.1173	-90333.2	180674.5	DLM
VAR(1)	3.8517	3.8395	–	–	Benchmark
Model 4	4.2049	4.0880	–	–	Optional

Model 2 is preferred by AIC, showing that modelling cross-station dependence matters for fit. Model 3 is slightly worse in AIC but more interpretable because dependence is tied to geographic distance. Predictive RMSE is similar across the three DLMs: cross-station structure improves the description of joint dynamics more than it improves one-step accuracy. The VAR(1) benchmark is not part of the required DLM comparison, but its lower RMSE is informative: it shows that short-run AR(1) persistence is the main predictive component missing from the random-walk DLMs.

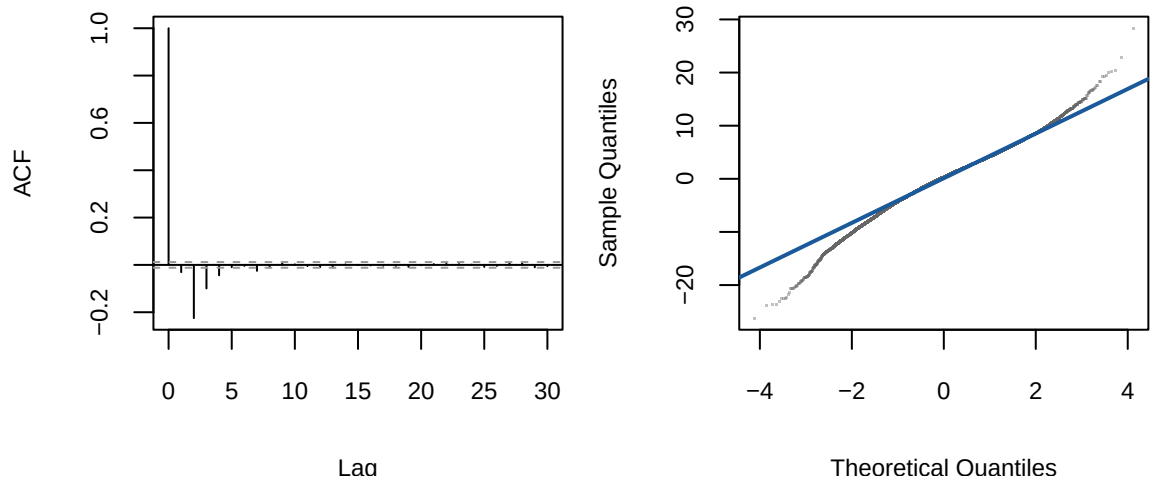


Figure 6. Residual diagnostics for Model 2: ACF and QQ plot of Houlton one-step prediction errors.

Figure 6 confirms this interpretation. The large lag-1 ACF indicates that the random-walk state equation does not fully capture daily persistence. This explains why the VAR(1) benchmark improves forecasting. The QQ plot is broadly linear in the center, but the tails are heavier than Gaussian, which is common for daily weather anomalies.

4.4 Discussion

The log-likelihood gain from Model 1 to Model 2 (1.4772×10^4 units) with only one additional parameter shows that the independence assumption is too restrictive. Model 1 treats the two latent station levels as

evolving separately, while Model 2 allows their innovations to be correlated. The estimated state-innovation correlation is 0.999, suggesting that the two stations are driven by an almost common regional temperature signal. This is consistent with the strong contemporaneous dependence observed in Figure 5.

Model 3 gives a more structured version of the same idea. Instead of estimating an unrestricted state covariance matrix, it links dependence to geographic distance through an exponential spatial correlation. The estimated spatial decay parameter is 9×10^{-4}, implying a spatial correlation of 0.9975. Since the two stations are geographically close, this near-perfect correlation is plausible. Thus, Model 2 provides the best likelihood-based fit, while Model 3 provides a more parsimonious and geographically interpretable representation of the same latent co-movement.

The predictive RMSEs are nevertheless very similar across the three DLMs. This does not contradict the likelihood results. The log-likelihood evaluates the joint probabilistic model, including whether the covariance structure between the two stations is well represented. RMSE, instead, evaluates only marginal one-step-ahead point forecast accuracy. Therefore, cross-station dependence can substantially improve the joint model fit without producing a large reduction in one-step prediction errors. In this application, daily forecasts remain dominated by local short-run variability.

The VAR(1) benchmark helps identify this remaining predictive gap. Its lower RMSE shows that yesterday's anomaly contains useful information for tomorrow's anomaly. The estimated own-lag coefficients are 0.437 for Houlton and 0.418 for Augusta, while the cross-lag effects are smaller, equal to 0.16 and 0.161. This suggests that short-run persistence is mainly within each station, whereas cross-station dependence is mostly contemporaneous and regional.

The VAR(1) should therefore be interpreted as a diagnostic benchmark rather than as a replacement for the DLM analysis. The VAR is optimized for short-run point prediction, but it does not provide the same latent-state interpretation as the DLM and does not separate observation noise from an evolving latent level. The comparison suggests a natural extension: a richer state-space model with AR(1) observation errors or a more persistent state transition equation.

Finally, Model 4, fitted on raw TMAX with a Fourier seasonal component and spatial covariance, gives RMSE values 4.20 and 4.09. This supports the adequacy of the simpler seasonal pre-adjustment used in Models 1–3: explicitly modelling the annual cycle does not materially improve one-step-ahead prediction in this two-station application.

4.5 Limitations and Extensions

Several limitations should be kept in mind. In the GISTEMP analysis, the HMM treats warming as a sequence of discrete states, while the physical process is likely smoother. The DLM is better suited to gradual change, but the selected random walk plus noise model still leaves some residual autocorrelation. A richer climate model could include seasonal components directly, time-varying variances, or autoregressive errors.

In the GHCN analysis, only two stations are used. This is sufficient to illustrate bivariate DLMs and spatial covariance ideas, but not enough to estimate a rich spatial dependence structure. With more stations, the spatial decay parameter in Model 3 would be identified from many pairwise distances, making the spatial interpretation more informative.

Finally, the prediction exercise focuses on one-step-ahead point forecasts. A fuller probabilistic comparison could also evaluate prediction intervals or log predictive scores, which would be especially relevant for weather applications where uncertainty quantification matters as much as point accuracy.

5. Conclusion

This analysis addressed two related questions: whether Northern Hemisphere temperature anomalies show a persistent long-run increase, and whether state-space models can also be used for short-run multivariate weather prediction.

For the GISTEMP monthly series, both the HMM and the selected random walk plus noise DLM provide consistent evidence of warming. The HMM describes warming as a progressive shift from cooler to warmer regimes: the warmest regime accounts for a share of 0 of months before 1951 and 0.707 after 1980. The DLM gives a complementary continuous picture: the smoothed latent level rises throughout the sample, and the average local yearly change increases from 0.0021 in 1880–1950 to 0.0283 in 1981–2025. The agreement

between a discrete-regime model and a continuous latent-level model supports the conclusion that the increase is persistent rather than driven by isolated extreme observations.

For the GHCN daily data, the bivariate DLMs show that nearby stations share a strong latent temperature signal. Model 2, with unrestricted state-innovation covariance, is preferred by AIC, while Model 3 provides a geographically interpretable spatial covariance with only modest loss of fit. Predictive RMSE is similar across the three DLMs, indicating that cross-station structure improves the joint probabilistic description more than marginal one-step point accuracy. The VAR(1) benchmark reveals that short-run autoregressive persistence is the main remaining predictive gap. Overall, state-space models provide a coherent framework for both long-run climate trend estimation and multivariate weather modelling, while also suggesting natural extensions such as AR(1) errors, richer seasonal structures, and larger spatial DLMs.

References

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